

MAP THE WHOLE UNIT ON ONE SCREEN — 28 SYLLABUS HEADS, 5 BLOCKS

History A-01 Bakewell → Lush → BLUP → genomics	Breeds A-02 Classification + Indian breed lists	Economic traits A-03 Dairy, meat, wool, egg	Selection B-04 Types & principles
Response B-05 $\Delta G = i \cdot h^2 \cdot \sigma_p / L$, factors	Bases B-06 Individual, pedigree, family, sib	Progeny test B-07 Combined & indirect selection	Methods B-08 Tandem · ICL · Selection index
Mating systems C-09 Full classification chart	Inbreeding F C-10 Wright's formula, relationship R	Consequences C-11 Depression & applications	Outbreeding C-12 Outcross, crossbreed, grading up
Heterosis C-13 Theories & % formula	Utilising it C-14 Rotational, terminal, synthetic	Combining ability C-15 GCA/SCA · RS & RRS	Dairy strategy D-16 Cattle & buffalo
Small stock D-17 Sheep, goat, swine	Poultry D-18 Egg & broiler lines	Sire evaluation D-19 CC → BLUP	ONBS D-20 Nucleus-multiplier-base
New breeds D-21 Synthetics & strains	Policies E-22 NPCBB, Rashtriya Gokul Mission	Conservation E-23 In-situ / ex-situ, AnGR	Biotech tools E-24 AI, MOET, MAS, genomic selection
Disease resistance E-25 Major genes, MAS	Pets F-26 Dog & cat breeds, pedigree sheet	Pet management F-27 Dogs, cats, pet birds	Wildlife F-28 N_e planned & controlled breeding

★ HIGHEST-YIELD IN THIS UNIT

- **Response to selection** — ΔG formula + all factors affecting it.
- **Bases of selection** — the 7-row comparison.
- **Tandem vs ICL vs Selection index** — sure differentiate.
- **Inbreeding** — F, consequences, depression, applications.
- **Heterosis** — definition, formula, theories, systems.
- **Sire evaluation** — contemporary comparison & BLUP.
- **ONBS** — diagram + advantages.
- **Crossbreeding strategy for Indian dairy cattle.**
- **Conservation of AnGR** — in-situ vs ex-situ.
- **Effective population size N_e** — formula + 50/500 rule.

1 HISTORY OF ANIMAL BREEDING — THE NAMES THEY ASK

Robert Bakewell (1725–95) — pioneer of breeding **practice**; Leicester sheep, Shire horse. Introduced **inbreeding + progeny testing** + “**breed the best to the best**”.
Colling brothers & Bates — Shorthorn; line breeding.
Sewall Wright (1921–22) — **inbreeding coefficient F**, path coefficients.
Fisher, Wright & Haldane — founders of **population & quantitative genetics**.
Jay L. LUSH (1937, “Animal Breeding Plans”) — **FATHER OF MODERN ANIMAL BREEDING**.

Hazel & Lush (1943) — **SELECTION INDEX**.
Comstock, Hess & Bell (1949) — **RRS**.
Robertson & Rendel (1954) — **Contemporary Comparison**.
C.R. Henderson (1949, 1975) — **BLUP** / animal model.
Meuwissen *et al.* (2001) — **GENOMIC SELECTION**.
India: KVS 1952 · ICDP 1964 · Operation Flood 1970 (NDDB, Kurien) · NPCBB 2000 · Rashtriya Gokul Mission 2014.

2 CLASSIFICATION OF BREEDS + INDIAN BREED LISTS ★ EASY MARKS

BREED = a group of animals of common origin with **distinguishable characteristics** (colour, horn, size, performance) that are **transmitted to the offspring**. **Strain** = sub-group developed by one breeder; **Variety** = differs in one visible character; **Type** = based on function, cuts across breeds.

Bases of classification: ① **Utility** (milk, draught, dual; mutton/wool; egg/broiler) ② **Origin** (indigenous · exotic · crossbred) ③ **Taxonomy** (*Bos indicus* — humped, heat & tick tolerant; *Bos taurus* — humpless, high yield; *Bubalus bubalis*) ④ **Size** ⑤ **Horn / coat pattern**.

Species	Key Indian breeds (write 4–5 each)
Cattle — Milch	Sahiwal, Red Sindhi, Gir, Tharparkar (Tharparkar also dual)
Cattle — Draught	Amritmahal, Hallikar, Khillari, Kangayam, Nagori, Malvi, Bachaur, Kherigarh
Cattle — Dual	Haryana, Ongole, Kankrej, Deoni, Rathi, Mewati, Dangi, Gaolao, Krishna Valley
Buffalo	Murrah (best milch, “black gold”), Nili-Ravi, Jaffarabadi (heaviest), Surti, Mehsana, Bhadawari (highest fat ~13%), Nagpuri, Pandharpuri, Toda, Banni
Sheep	Carpet wool — Chokla (best Indian wool), Nali, Magra, Marwari ; Mutton — Nellore, Mandya, Deccani ; Apparel wool — Gaddi, Bhakarwal, Rampur Bushair
Goat	Jamunapari (largest, “Pari”), Beetal, Barbari, Sirohi, Marwari, Osmanabadi, Black Bengal (best skin, prolific), Changthangi (pashmina), Malabari, Surti
Pig	Indigenous — Ghungroo, Ankamali, Niang Megha, Agonda Goan, Doom ; Exotic — Large White Yorkshire (most used), Landrace, Hampshire, Duroc
Poultry	Native — Aseel (game), Kadaknath (black meat), Chittagong, Busra, Ghagus, Nicobari ; Improved — Vanaraja, Gramapriya, Krishibro, Giriraja ; Exotic — White Leghorn (egg), Cornish & Plymouth Rock (broiler), RIR

TRAP

NBAGR, Karnal is the national authority that **registers** Indian breeds. India has ~50+ cattle and ~20 buffalo registered breeds. **Zebu (*Bos indicus*) traits** — hump, dewlap, heat & tick tolerance, disease resistance, low yield, late maturity; **taurine** — no hump, high yield, early maturity, heat-sensitive.

3 ECONOMIC TRAITS OF LIVESTOCK & POULTRY AND THEIR IMPORTANCE

Enterprise	Economic traits (the selection objectives)	Typical h^2 & note
Dairy	Total/305-day milk yield, fat % & SNF %, lactation length (305 d), peak yield & persistency, age at first calving (AFC), calving interval (12–13 mo), dry period (60 d), service period, milking speed, udder & teat conformation, longevity/herd life, feed efficiency, mastitis & SCC, temperament	Milk yield 0.25–0.30 ; fat % 0.5–0.6 ; AFC & calving interval < 0.10 (manage, don't select)
Meat / growth	Birth weight, weaning weight, average daily gain (ADG), feed conversion ratio (FCR), dressing %, carcass length, backfat thickness, eye-muscle area, marbling, meat : bone ratio	Growth & carcass traits 0.30–0.60 → respond well to individual selection
Sheep — wool	Greasy & clean fleece yield, staple length, fibre diameter (micron), medullation %, crimp, lustre, yield %, lambing % & twinning	Fleece weight 0.3–0.4 ; fibre diameter 0.4–0.5 ; lambing % < 0.10
Swine	Litter size at birth & at weaning, birth & weaning weight, number of functional teats, ADG, FCR, backfat, dressing %, age at 90 kg	Litter size 0.10–0.15 (low, use crossbreeding); backfat 0.4–0.5
Poultry — egg	Age at sexual maturity (ASM), egg number to 40/72 weeks, egg weight, shell thickness, Haugh unit, fertility, hatchability, FCR, livability (broodiness is undesirable)	Egg number 0.2–0.3 ; egg weight 0.5–0.6 ; fertility & hatchability < 0.15
Poultry — broiler	6-week body weight, FCR, breast yield, livability, feathering rate, leg soundness	6-wk body weight 0.35–0.45

WHY “ECONOMIC” TRAITS — THE ANSWER THEY WANT

A trait is economic when a unit change in it changes **net profit**; the objective therefore combines traits by their **relative economic weights** (the basis of the **selection index**).
Universal pattern: **production & carcass traits have HIGH h^2 (select on the individual); reproduction, fertility and viability traits have LOW h^2 (improve by crossbreeding and management).**

4 SELECTION — TYPES & RESPONSE ★★

SELECTION = allowing certain animals to be the **parents of the next generation** more often than others — i.e. **differential reproduction**. It **does not create new genes**; it only **changes gene frequencies** by increasing the frequency of favourable alleles. **Selection + culling are two sides of the same act.**

Type	Effect on the population
Natural vs Artificial	Nature favours fitness ; man favours economic merit — the two often oppose each other.
Directional	Favours one extreme → the mean shifts . The normal breeding situation (higher milk, faster growth).
Stabilising (normalising)	Favours the intermediate , culls both extremes → variance falls , mean unchanged. Ex: egg weight, birth weight (dystocia at both ends).
Disruptive (diversifying)	Favours both extremes → population splits into two groups; used to create divergent lines.

Selection differential $S = (\text{mean of selected parents}) - (\text{mean of the population})$

Intensity of selection $i = S / \sigma_p$ (a standardised, unit-free S)

RESPONSE per generation $R = \Delta G = h^2 \times S = i \cdot h^2 \cdot \sigma_p = i \cdot r_{IA} \cdot \sigma_A$

RESPONSE per YEAR $\Delta G/\text{yr} = (i \cdot h^2 \cdot \sigma_p) / L$ (L = generation interval)

Realised heritability $h^2 = R / S$

Generation interval L = average age of the parents when their replacement offspring are born. **Cattle 4.5–6 yr • Buffalo 6–8 • Sheep & goat 3–4 • Pig 2 • Poultry 1–1.5 yr.** This is why **poultry improve fastest and buffalo slowest**.

5 FACTORS AFFECTING RESPONSE TO SELECTION ★ WRITE ALL EIGHT

- Heritability of the trait** — the higher h^2 , the more of the observed superiority is transmitted. Low- h^2 traits respond poorly to individual selection.
- Selection differential / intensity (i)** — set by the **proportion selected**, hence by the species' **reproductive rate** and **culling rate**. Poultry and pigs allow high intensity; cattle very little on the female side.
- Phenotypic variability (σ_p)** — no variation, no selection. A uniform herd offers little scope.
- Generation interval (L)** — yearly response is **inversely** proportional to L; shortening L (young sires, MOET, genomic selection) is often the fastest route to gain.
- Accuracy (r_{IA})** — how well the estimated breeding value reflects the true one; raised by **more records, more progeny, BLUP and genomic data**.
- Number of traits selected for** — gain per trait falls roughly as $1/\sqrt{n}$. Selecting for 4 traits halves the progress in each.
- Genetic correlations** — a favourable r_A gives free correlated gain; an **antagonistic** one (milk yield vs fertility, growth vs feed efficiency) slows or reverses progress.
- Inbreeding, G × E and environment** — inbreeding depresses performance; a poor environment masks genetic differences and lowers h^2 ; a **selection plateau** is reached when additive variance is exhausted.

EXAM TRAP

Response is per generation; gain is quoted per YEAR — always divide by L. The **four "levers"** in $\Delta G/\text{yr} = i \cdot r_{IA} \cdot \sigma_A / L$ are the only ways a plan can work: raise **i**, raise **accuracy r**, exploit **σ_A** , or cut **L**. Genomic selection wins by **cutting L and raising r in young animals**.

6 BASES OF SELECTION — WHICH INFORMATION DO WE USE? ★★ CERTAIN QUESTION

Basis	Information used	Best used when	Limitation
INDIVIDUAL / MASS / phenotypic	The animal's own performance record	h^2 HIGH (>0.4) ; trait expressed early and in both sexes ; cheap & simple. Ex: body weight, growth rate, fleece weight, egg weight	Useless for sex-limited traits, carcass traits, and low- h^2 traits
PEDIGREE	Records of ancestors (sire, dam, grandparents)	Animal is too young to have a record; trait is sex-limited or measurable only after slaughter; low h^2 ; choosing a bull calf	Contribution halves each generation ($1/2$, $1/4$, $1/8$) — remote ancestors are almost worthless ; ancestors reared in a different era/environment
FAMILY (between-family)	Mean of the whole family ; family accepted or rejected as a unit	LOW h^2 and LARGE families (poultry, pigs); family mean averages out environment	Fails if members share a strong common environment ; ↑ inbreeding
WITHIN-FAMILY	Best individuals inside each family	Common environmental effect is large (litter/maternal effects)	Losses between-family genetic differences
SIB (full-sib / half-sib)	Records of brothers & sisters	Sex-limited traits and destructive measurements (carcass, meat quality); a young cockerel judged on his sisters' egg records	Full-sib data inflated by common environment & dominance
PROGENY TESTING	Performance of the animal's OFFSPRING	The most accurate method for SIREs ; essential for sex-limited traits (milk, eggs) and for low-h^2 traits; the basis of bull selection in AI	Long generation interval, costly, few sires can be tested , needs large progeny groups & good recording
COMBINED	Own + pedigree + sib + progeny, each weighted by its accuracy	The most efficient of all — this is exactly what BLUP / animal model does	Needs complete pedigree & computing power
INDIRECT	Selection on a correlated trait	Target trait is hard, late, expensive or sex-limited; ex: SCC for mastitis resistance , backfat by ultrasound for carcass fat	Only pays when $r_A \cdot h_{\text{secondary}} > h_{\text{primary}}$

Accuracy of progeny test $r = \sqrt{[n / (n + k)]}$, $k = (4 - h^2)/h^2$

ONE-LINE RULE TO QUOTE

High h^2 → individual selection. Low h^2 → family, sib or progeny testing. Sex-limited → progeny testing (sires) or sib/pedigree. Destructive traits → sib selection. Everything at once → BLUP (combined).

7 METHODS OF SELECTION FOR SEVERAL TRAITS — SINGLE VS MULTI-TRAIT ★ SURE DIFFERENTIATE

Method	How it works	Merit	Demerit
TANDEM	Select for ONE trait at a time until it reaches the desired level, then move to the next	Simple; full intensity on one trait; fastest gain in that one trait	SLOWEST overall ; earlier trait slips back ; ignores genetic correlations; wastes generations
INDEPENDENT CULLING LEVEL (ICL)	A minimum standard is fixed for EVERY trait ; an animal failing any one is culled, however good elsewhere	Selects for all traits simultaneously; can cull early at different ages (cheap, practical on farm)	Discards an outstanding animal for one small fault ; does not allow merit in one trait to compensate a deficiency in another
SELECTION INDEX (Hazel & Lush, 1943)	All traits combined into a SINGLE SCORE weighted by their heritabilities, genetic & phenotypic correlations and relative economic values ; animals ranked on the index	THEORETICALLY THE MOST EFFICIENT — allows compensation between traits, uses all information, maximises total economic gain	Needs reliable estimates of h^2 , r_A , r_p and economic weights; all traits must be recorded on every animal before culling; computationally heavy

Index $I = b_1P_1 + b_2P_2 + b_3P_3 + \dots + b_nP_n$

P = phenotypic record of each trait; **b** = **partial regression weight** derived from $Pb = Gv$, where v = relative economic values. **Aggregate genotype** $H = a_1A_1 + a_2A_2 + \dots$; the index is built to maximise the correlation r_{IH} between I and H.

EFFICIENCY ORDER — ONE LINE FOR THE EXAM

SELECTION INDEX > INDEPENDENT CULLING LEVEL > TANDEM. In practice a **combination** is used: cull grossly on ICL for cheap early traits (survival, conformation, gross defects), then rank the survivors on a **selection index**.

INDIAN DAIRY EXAMPLE

A bull-mother index for Sahiwal might weight **first-lactation 305-day yield, fat %, age at first calving and calving interval**. The **National Dairy Plan progeny-testing projects** now rank bulls by **BLUP-based total merit indices** — selection indices computed with an animal model.

TRAP

The **selection index and BLUP EBV are not opposites** — BLUP produces the breeding values that a modern index combines. Also remember **ΔG per trait falls as $1/\sqrt{n}$** , so an index with **too many traits** improves none of them appreciably: keep the objective to a **few economically important traits**.

8 CLASSIFICATION OF MATING SYSTEMS — DRAW THIS CHART

MATING SYSTEMS — how selected animals are paired

A. On GENETIC RELATIONSHIP

INBREEDING — related animals

- **Close breeding** (full-sib, parent-offspring)
- **Line breeding** (keeps kinship to one outstanding ancestor high, total F low)

OUTBREEDING — unrelated animals

- **Outcrossing** (unrelated, same breed)
- **Crossbreeding** (different breeds, same species)
- **Species / genus hybridisation** (mule, dzo)
- **Grading up** (repeated purebred sires on native stock)

C. RANDOM MATING (panmixis) — every male has an equal chance of mating any female; the reference point of the Hardy-Weinberg law

B. On PHENOTYPIC RESEMBLANCE

ASSORTATIVE (like × like)

Positive — best × best; ↑ uniformity, ↓ variance between groups

DISASSORTATIVE (unlike × unlike)Also called **corrective mating** — a fault in one parent is compensated by the other; ↑ heterozygosity

9 INBREEDING — COEFFICIENT F & RELATIONSHIP R ★★

Inbreeding = mating of individuals **more closely related to each other than the average of the population** from which they come.

$$\text{Wright's } F_{X} = \sum \left(\frac{1}{2} \right)^{(n_1 + n_2 + 1)} (1 + F_A)$$

n_1, n_2 = number of generations from sire and from dam back to the **common ancestor** A; F_A = inbreeding of that ancestor; Σ = sum over **every path** through **every common ancestor**.

Definition of F: the **probability that the two alleles at a locus in an individual are IDENTICAL BY DESCENT** (i.e. replicas of one allele in a common ancestor). Range **0 → 1**.

$$\text{Coefficient of relationship } R_{XY} = 2F_{XY} / \sqrt{(1+F_X)(1+F_Y)}$$

Mating / relationship	F of progeny	R between the pair
Full sibs (brother × sister)	0.25	0.50
Parent × offspring	0.25	0.50
Half sibs	0.125	0.25
Grandparent × grandoffspring	0.125	0.25
First cousins	0.0625	0.125
Uncle/aunt × niece/nephew	0.125	0.25

$$\text{Rate of inbreeding } \Delta F = 1/(8N_m) + 1/(8N_f) = 1/(2N_e)$$

Practical rule: keep ΔF below 1% per generation ($\leq 0.5\%$ is safer).

10 CONSEQUENCES OF INBREEDING & ITS APPLICATIONS

GENETIC CONSEQUENCES:

- ↑ **HOMOZYGOSITY** and ↓ **heterozygosity** — heterozygotes fall by the factor $(1 - F)$: $AA = p^2 + pqF$; $Aa = 2pq(1 - F)$; $aa = q^2 + pqF$.
- **Gene frequency is NOT changed** — only the genotypic frequencies are redistributed.
- **Exposes deleterious recessives** in the homozygous state → hidden genetic defects appear.
- Splits the population into **inbred lines**: variance **WITHIN** lines falls, **BETWEEN** lines rises.
- Increases **uniformity and PREPOTENCY** (a sire's ability to stamp his characters on his progeny).

PHENOTYPIC CONSEQUENCE — INBREEDING DEPRESSION: the decline in mean performance for traits associated with **fitness**. It is **roughly LINEAR** with **F** and is greatest in traits with **low h^2** and **high dominance variance**: **fertility, conception rate, litter size, viability & survivability, growth rate, milk yield**. Traits with mainly additive variance (fat %, skeletal measurements, fibre diameter) are hardly affected. **Cause:** expression of homozygous deleterious recessives (**dominance theory**) and loss of heterozygote advantage (**overdominance**) — it is simply the **reverse of heterosis**.

APPLICATIONS: ① develop **INBRED LINES** for later crossing to exploit heterosis (poultry, swine, lab animals) ② **fix and make uniform** a desirable type — how **every breed was originally formed** ③ **detect carriers** of undesirable recessives ④ raise **prepotency** of elite sires ⑤ produce **isogenic lab strains** ⑥ line breeding to conserve an outstanding ancestor's genes.

TRAP

Line breeding is a MILD form of inbreeding, not a separate system. Inbreeding itself never improves a herd — it only exposes what is already there, and is safe only with rigorous simultaneous selection and culling.

11 OUTBREEDING • ITS FORMS & CONSEQUENCES

Form	Definition & use
Outcrossing	Mating of unrelated animals WITHIN the same breed (no common ancestor for 4–6 generations). Restores vigour in an inbred purebred herd without losing breed identity.
Crossbreeding	Mating animals of two different breeds of the same species. Produces heterosis and combines complementary breed strengths — the mainstay of Indian dairy, pig and poultry production.
Species / genus hybridisation	Crossing different species : horse ♀ × ass ♂ = MULE ($2n = 63$, sterile, great draught vigour); ass ♀ × horse ♂ = hinny ; yak × cattle = dzo . Hybrids are usually sterile — chromosome numbers differ.
GRADING UP	Repeated use of purebred sires of one breed on a nondescript/native female population , generation after generation. Exotic inheritance: F_1 50% → 75% → 87.5% → 93.75% → 96.87% . In 4–6 generations the graded animals are practically purebred. Cheapest and quickest way to upgrade a large nondescript population (e.g. grading up nondescript buffaloes with Murrah bulls).

CONSEQUENCES: ↑ **heterozygosity** and ↓ **homozygosity** · **masks deleterious recessives** · **improves fertility, viability and vigour (heterosis)** · combines the merits of two breeds (**complementarity**) · **gene frequency unchanged in the population as a whole**. **Drawbacks:** loss of uniformity and of breed purity; **heterosis is NOT fixable — it halves in F_2 (recombination loss)**; crossbreds may be poorly adapted; needs continuous supply of purebred parent stock; indiscriminate crossbreeding **erodes indigenous germplasm**.

12 HETEROSIS (HYBRID VIGOUR) — SHULL, 1914 ★★

Definition: the **superiority of the crossbred offspring over the AVERAGE of its two parental breeds** (mid-parent value) for a given trait. Its opposite is inbreeding depression.

$$\text{Heterosis \% } H\% = \left[\frac{(F_1 \text{ mean} - \text{Mid-parent mean})}{\text{Mid-parent mean}} \right] \times 100$$

Genetic basis $H \propto \Sigma d y^2$ (d = dominance deviation, y = difference in gene frequency between the breeds)

- **THEORIES:** ① **Dominance** — each parent's dominant favourable alleles **mask the other's deleterious recessives** (most accepted). ② **Overdominance** — the heterozygote **Aa is itself superior**. ③ **Epistasis** — favourable between-locus interactions.
- **Two requirements:** the trait must show **DOMINANCE** ($d \neq 0$) and the two breeds must **DIFFER IN GENE FREQUENCY** ($y \neq 0$). Hence heterosis is **largest between genetically distant breeds**.
- **Greatest in LOW- h^2 fitness traits** — fertility, litter size, viability, longevity (10–30%); **small in high- h^2 traits** — fat %, fibre diameter, body measurements.
- **Types:** **individual** (in the crossbred itself), **maternal** (from a crossbred dam — mothering ability, milk), **paternal** (from a crossbred sire — libido, semen).
- **NOT FIXABLE:** in F_2 heterosis **halves** with heterozygosity (recombination loss) — crossbreds must be **regenerated each generation** or stabilised into a synthetic.

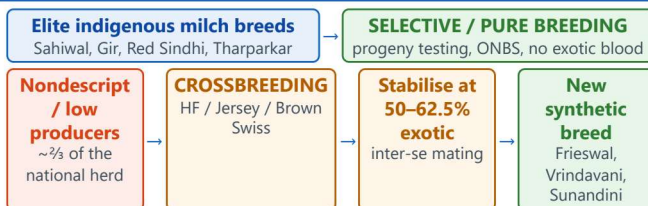
13 SYSTEMS OF UTILISING HETEROSIS + COMBINING ABILITY (RS & RRS)

System	How it works / where used
Two-breed (simple) cross	$A \times B \rightarrow F_1$ all marketed. Maximum individual heterosis, no maternal heterosis .
Three-breed cross	$(A \times B) \times C$ ♂ — gains individual + maternal heterosis. Common in pigs .
Criss-cross / rotational (2-breed)	Crossbred females bred alternately to A and B sires; retains $\approx$ 67% heterosis , replacements bred on the farm.
Rotational 3-breed	Sires of A, B, C used in rotation; retains $\approx$ 86% heterosis .
Terminal (static terminal) crossing	All crossbred progeny go for slaughter ; females bought in. Maximum heterosis + complementarity; no replacements produced .
Roto-terminal	Rotation to breed replacements + a terminal sire on the older females. Best of both.
SYNTHETIC / COMPOSITE breed	Crossbreds inter-se mated and stabilised into a new breed — retains part of the heterosis permanently . Karan Swiss, Karan Fries, Frieswal, Sunandini, Vrindavani, Taylor, Jersind, Avikalin, Bharat Merino, Hissardale, Vanaraja, Gramapriya .
Topcross · Incross · Incrossbred	Inbred sire × non-inbred dams; crosses of inbred lines of the same or different breeds — poultry .

COMBINING ABILITY: **GCA** = the **average performance of a line in all its crosses** → due to **ADDITIVE** gene action. **SCA** = the **deviation of one particular cross** from what its parents' GCA predicted → due to **DOMINANCE & EPISTASIS**.

RS (Recurrent Selection) — one line is selected on its cross performance with a **constant inbred tester**; improves **GCA** of that line only. **RRS (Reciprocal Recurrent Selection — Comstock, Hess & Bell, 1949)** — **two lines are each used as the tester for the other**, and both are selected and improved simultaneously; improves **BOTH GCA and SCA**. RRS is the standard method in **commercial poultry (and pig) line development**.

14 BREEDING STRATEGY — DAIRY CATTLE & BUFFALO ★★



- ▶ **Cattle** — the accepted national policy: ① Selection & pure breeding in the recognised indigenous milch breeds in their breeding tracts ② Selection for draught + conservation in draught breeds ③ Crossbreeding of nondescript cows with exotic dairy breeds, holding exotic inheritance at 50% (up to 62.5%), then inter-se mating of the crossbreds ④ Grading up of nondescript stock with the best-adapted indigenous breed ⑤ Country-wide AI with progeny-tested frozen semen, pedigree & milk recording.
- ▶ **Choice of exotic breed:** Jersey for hot, humid, low-input areas (small, early maturing, high fat, easy calving); Holstein-Friesian for cooler, well-managed, high-input areas (highest yield, low fat); Brown Swiss intermediate and hardy.
- ▶ **BUFFALO** — NO CROSSBREEDING with exotic. There is no superior exotic buffalo, so the strategy is: selection and pure breeding within Murrah, Nili-Ravi, Jaffarabadi, Surti, Mehsana etc., progeny testing of Murrah bulls (CIRB Hisar, NDRI, Network Project on Buffalo Improvement), and GRADING UP of nondescript buffaloes with Murrah — the single most successful Indian breeding programme. Constraints: late maturity, long calving interval, seasonality, low reproductive rate, poor heat detection (silent heat).
- ▶ **Selection criteria for bull mothers:** high first-lactation 305-day yield, fat %, early AFC, short calving interval, sound udder, freedom from genetic defects, good pedigree.

16 SIRE EVALUATION ★★ — WHY, AND THE METHODS

Why evaluate sires? A bull produces **no milk himself** (milk is a sex-limited trait), yet through AI one bull can sire **thousands** of daughters. **Over 75–90% of the genetic improvement in a dairy population comes through the sire path.** He must therefore be evaluated on his **daughters**.

Method	Basis & limitation
Dam–daughter comparison	Daughters' average vs their dams' average. Simple but ignores environmental & year differences and regression to the mean.
Simple daughter average	Only the daughters' mean — badly biased by herd and management.
Equal-parent index	$SI = 2\bar{D} - \text{Dam average}$ (assumes both parents contribute equally).
CONTEMPORARY COMPARISON (Robertson & Rendel, 1954)	Each daughter is compared with the contemporaries — unrelated herdmates of the same herd, year and season calving at the same time. Removes herd-year-season environmental effects ; the standard field method. Weighted by number of daughters.
Herdmate comparison (USDA)	A refinement of CC using a wider herdmate base.
Least-squares (Harvey)	Handles unequal and disproportionate subclass numbers ; fits fixed effects simultaneously.
★ BLUP (Henderson, 1949/1975)	Best Linear Unbiased Prediction. A mixed-model method that simultaneously estimates the fixed environmental effects and predicts the breeding values . It accounts for all genetic relationships among animals, unequal numbers, non-random mating, culling and genetic trend. Best = minimum error variance • Linear • Unbiased • Prediction of random effects. The ANIMAL MODEL form ranks every animal — cows and bulls — on one scale. Now universally used; combined with SNP data as genomic BLUP / ssGBLUP .

Accuracy $r = \sqrt{\frac{n}{n+k}}$, $k = \frac{(4-h^2)}{h^2} \rightarrow$ more daughters, more accuracy

Requirements for a valid progeny test: random allotment of semen across herds, **at least 30–50 daughters in 10+ herds**, accurate pedigree & milk recording, contemporaries available, and correction for **age, season, lactation number and lactation length (305-day, 2× milking, mature-equivalent)**.

15 SHEEP • GOAT • SWINE • POULTRY STRATEGIES

Species	Strategy & breeds used
SHEEP — wool	Crossbreeding of native ewes with Rambouillet, Merino, Corriedale, Russian Merino to improve fineness & fleece weight, then inter-se mating $\rightarrow$ Hissardale (first Indian crossbred sheep) , Avikalin, Avivastra, Bharat Merino, Nilgiri Synthetic . Select for greasy fleece weight, staple length, fibre diameter, low medullation .
SHEEP — mutton	Selection within Nellore, Mandya, Deccani, Marwari; crossing with Dorset, Suffolk for growth. Select for 6-month body weight, ADG, lambing % & twinning .
GOAT	Selection within Jamunapari, Beetal, Barbari, Sirohi (milk & meat) and Black Bengal (meat & superior skin, prolific). Crossing with Alpine / Saanen for milk, Boer for meat; Changthangi for pashmina. Traits: litter size, kidding interval, 6-month weight, milk yield.
SWINE	Crossbreeding of indigenous sows with Large White Yorkshire (most widely used), Landrace, Hampshire, Duroc — usually a three-breed or terminal cross to capture individual + maternal heterosis. Select for litter size at birth & weaning, ADG, FCR, backfat thickness, number of functional teats . Very high reproductive rate $\rightarrow$ very high selection intensity and short L (2 yr) .
POULTRY	Highly industrialised, separate egg-type and broiler lines . Egg: White Leghorn-based; select for ASM, egg number to 72 wk, egg weight, FCR, livability . Broiler: Cornish sire line × Plymouth Rock dam line ; select for 6-week body weight, FCR, breast yield . Methods: development of inbred and pure lines $\rightarrow$ strain crossing $\rightarrow$ RRS $\rightarrow$ 3- and 4-way commercial hybrids , with random-bred controls to measure genetic trend. Rural/backyard: dual-purpose coloured birds — Vanaraja, Gramapriya, Krishibro, Giriraja, CARI-Nirbheek .

17 OPEN NUCLEUS BREEDING SYSTEM (ONBS) + NEW BREED DEVELOPMENT

ONBS = a breeding structure in which the **genetically best animals of the whole population are gathered into a small NUCLEUS herd/flock** where intensive recording and selection are done; superior genes then flow down to the base. The nucleus is **OPEN** because **superior females (and males) from the base population are continuously allowed to migrate INTO the nucleus** (typically about **half** of nucleus replacements).



- ▶ **Advantages:** much **higher selection intensity** (choose from the whole population, not one herd) • **shorter generation interval** • **greater accuracy** (recording is feasible in a small nucleus) • **faster genetic gain**, especially where field recording is poor — ideal for **Indian village conditions** • **less inbreeding than a CLOSED nucleus** because of continuous inflow • rapid dissemination through **AI and MOET**.

- ▶ **Closed nucleus** = no inflow $\rightarrow$ higher intensity but rising inbreeding and loss of variability. **MOET-ONBS** multiplies the female contribution and is used in Indian **buffalo, sheep and goat** improvement.

DEVELOPMENT OF NEW BREEDS / STRAINS — steps: ① fix the **breeding objective** and the target environment ② choose **foundation breeds** that complement each other ③ **crossbreed** to the desired level of inheritance ④ **INTER-SE MATING** of the crossbreds ⑤ **rigorous selection** and culling for the objective over several generations ⑥ **close the flock/herd** and stabilise type ⑦ evaluate, expand, **register (NBAGR)** and disseminate.

INDIAN SYNTHETICS — QUOTE THESE

Cattle: **Karan Swiss** (Brown Swiss × Sahiwal/Red Sindhi), **Karan Fries** (HF × Tharparkar), **Frieswal** (HF × Sahiwal, 62.5% HF — ICAR/Army), **Sunandini** (Kerala), **Vrindavani** (IVRI — HF/BS/Jersey × Harijana), **Phule Triveni, Taylor, Jersind**. **Sheep:** **Hissardale, Avikalin, Avivastra, Bharat Merino, Nilgiri Synthetic**. **Poultry:** **Vanaraja, Gramapriya, Krishibro, Giriraja, CARI strains**.

18 CURRENT BREEDING POLICIES & PROGRAMMES — STATE AND COUNTRY

A **breeding policy** is a written government decision stating, **for each region and species: which breed is to be used, on which female population, by what method (selection / grading up / crossbreeding) and up to what level of exotic inheritance.** It is framed by the State Animal Husbandry Department within the national framework, because the breeding tract, agro-climate and farmer resources differ from state to state.

Programme	Year & purpose
Key Village Scheme (KVS)	1952 — first organised cattle-improvement scheme; village-level breeding units with selected bulls.
Intensive Cattle Development Project (ICDP)	1964 — package approach: breeding + feeding + health + marketing around milk-shed areas.
OPERATION FLOOD	1970, NDDB, Dr V. Kurien — the White Revolution; Anand-pattern dairy co-operatives; made India the world's largest milk producer.
NPCBB	2000 — National Project for Cattle & Buffalo Breeding: AI delivery at the farmer's door, quality frozen semen, breed conservation.
National Dairy Plan-I	2012 — scientific progeny testing & pedigree selection , ration balancing, village-based milk procurement.
RASHTRIYA GOKUL MISSION	2014 ★ — development & conservation of INDIGENOUS breeds ; Gokul Grams, National Kamdhenu Breeding Centres, Gopal Ratna & Kamdhenu awards, e-Pashuhaat portal, Pashu Sanjivani & animal UID (INAPH).
Nationwide AI Programme (NAIP) · NPBBDD · NLM	Free AI coverage in low-performing districts; National Livestock Mission for small ruminants, pigs and poultry.

Institutions to name: NBAGR Karnal (registration & characterisation of breeds) · NDDB Anand · NDRI Karnal · CIRB Hisar (buffalo) · CSWRI Avikanagar (sheep) · CIRG Makhdoom (goat) · CARI & IVRI Izatnagar · CPDOs · Central Frozen Semen Production & Training Institute (Hessarghatta) · State Livestock Development Boards · Central Herd Registration Scheme.

19 CONSERVATION OF ANIMAL GENETIC RESOURCES (ANGR) ★★

Conservation = all human activities, including strategies and plans, that ensure the **continued existence, use and availability** of animal genetic resources.

Preservation is the narrower act of **holding the material unchanged** (usually frozen).

WHY conserve? insurance against unknown future needs (climate, disease, market) · unique **adaptation, heat & tick tolerance, disease resistance and draught ability** of indigenous breeds · they are the raw material for **future selection and crossbreeding** · **scientific, cultural and socio-economic** value to rural livelihoods · **ethical duty**; a breed lost is lost forever.

CAUSES of genetic erosion: **indiscriminate crossbreeding** and unregulated grading up · concentration on a **few high-yielding exotic breeds** · **mechanisation** replacing draught cattle · loss of **common grazing land** and changed farming systems · small population size → **drift and inbreeding** · disease outbreaks, drought and disasters · lack of economic incentive to the keeper.

Method	What it means
IN-SITU (in the native habitat)	Maintaining live populations in their own breeding tract under the farmer's management: breed societies & breeders' associations, herd/flock registration, subsidy and incentive schemes, elite bull production, gaushalas, Gokul Grams, protected breeding tracts, herd books, community/village-level conservation, providing an economic market for the breed's product. Advantage: the breed keeps evolving with its environment and stays useful.
EX-SITU IN-VIVO	Live animals maintained outside the breeding tract — government farms, ICAR institute herds, university farms, zoos and rescue centres.
EX-SITU IN-VITRO (cryoconservation)	Cryopreservation in liquid nitrogen (–196 °C) of semen, embryos, oocytes, somatic cells and DNA in a gene/germplasm bank. Cheap, safe, indefinite; but the breed stops evolving and cannot be used <i>in situ</i> .

Supporting tools: characterisation and registration by NBAGR; **FAO DAD-IS** global database; breed risk-status classes (**extinct, critical, endangered, vulnerable, at risk, not at risk**) judged on **number of breeding females, N_e and population trend**; molecular **microsatellite/SNP** characterisation to prioritise genetically distinct breeds; **Minimum Viable Population** and $N_e \geq 50$ targets.

20 REPRODUCTIVE & BIOTECHNOLOGICAL TOOLS FOR GENETIC IMPROVEMENT

Tool	Genetic contribution — which term of $\Delta G = i \cdot r \cdot \sigma_A / L$ it improves
ARTIFICIAL INSEMINATION (AI) + frozen semen	Hugely raises selection intensity on the sire path (one bull → thousands of daughters), permits progeny testing , breaks the size barrier, controls venereal disease, economical. Limits: needs trained inseminators & heat detection; can spread a genetic defect rapidly and raise population inbreeding.
Sexed semen (flow cytometry, ~90%)	More heifer calves from the best cows → raises female selection intensity ; fewer unwanted male calves.
Oestrus synchronisation	Fixed-time AI, better records, group management.
MOET (Multiple Ovulation & Embryo Transfer)	Multiplies the DAM's contribution (an elite cow can give many calves a year) and shortens the generation interval — the basis of MOET nucleus schemes.
IVF / IVEP · OPU	Embryos from slaughterhouse or live-donor oocytes; large numbers, cheaper than MOET.
Cloning (SCNT)	Exact genetic copies of elite animals — NDRI buffalo clones Garima, Samrupa. Costly, low efficiency.
Transgenesis & gene editing (CRISPR-Cas9)	Introducing/editing specific genes — polledness, disease resistance, milk composition.
MAS & QTL mapping	Selection on DNA markers linked to a QTL — useful for low-h^2, sex-limited, late-expressed or carcass traits and for carrier detection (BLAD, CVM, DUMPS).
★ GENOMIC SELECTION (Meuwissen <i>et al.</i> , 2001)	Thousands of SNPs across the whole genome are used to compute a GEBV in a reference population. Doubles the rate of genetic gain mainly by giving useful accuracy in very young animals , so bulls can be used at 1–2 years instead of 6 — it halves the generation interval and can largely replace conventional progeny testing.

21 BREEDING FOR DISEASE RESISTANCE

Resistance = ability of the animal to **limit establishment/multiplication of the pathogen.** **Tolerance** = ability to remain productive **despite** infection.

Resilience = ability to recover after challenge.

APPROACHES: ① **Direct selection** on the disease trait (incidence, clinical score) ② **Indirect selection** on a correlated indicator — e.g. **somatic cell count (SCC) for mastitis** (h^2 0.10–0.15 vs mastitis incidence 0.02–0.05), **FAMACHA/FEC** for worm resistance ③ **Major genes and MAS** ④ **crossbreeding** to introduce a resistant breed's genes ⑤ **transgenesis / gene editing.**

Gene / breed	Resistance
BoLA (MHC) haplotypes	Mastitis, general immune responsiveness in cattle
NRAMP1 (SLC11A1)	Brucellosis & tuberculosis resistance in cattle
PRNP genotype — the ARR allele	Scrapie resistance in sheep (basis of national ram-genotyping schemes)
Mx gene	Influenza / viral resistance in poultry
F18 / K88 (F4) receptor	E. coli diarrhoea & oedema disease resistance in pigs
N'Dama cattle	Trypanotolerance
Bos indicus (zebu)	Tick and tick-borne disease resistance, heat tolerance
Naked-neck & frizzle genes	Heat tolerance in tropical poultry

Advantages: the improvement is **permanent, cumulative and inherited**; **no drug residues or resistance**; cheaper than lifelong treatment; helps where vaccines are unavailable. **Limitations:** **very low h^2** for most diseases; **needs exposure/challenge** to record the trait; **often antagonistic to production** traits; pathogens **evolve**; resistance to one disease may not extend to others; long time and large recorded populations required.

22 DOG & CAT BREEDS · PEDIGREE SHEET · BREED TRAITS

DOG — Kennel-Club groups (7): ① **Sporting/Gun** (Labrador, Golden Retriever, Pointer, Cocker Spaniel) ② **Hounds** — sight hounds hunt by sight, scent hounds by smell (Greyhound, Beagle, Dachshund, **Rampur & Mudhol Hound**) ③ **Working** (Boxer, Rottweiler, Doberman, Great Dane, Mastiff) ④ **Terriers** (Bull & Fox Terrier) ⑤ **Toy** (Pug, Spitz, Chihuahua) ⑥ **Non-sporting/Utility** (Dalmatian, Bulldog, Poodle) ⑦ **Herdling** (German Shepherd, Collie, **Gaddi & Bakharwal**).

Indian dog breeds: **Rajapalayam** (white, Tamil Nadu), **Chippiparai**, **Kanni**, **Kombai**, **Pandikona**, **Mudhol/Caravan Hound**, **Rampur Hound**, **Gaddi**, **Bakharwal**, **Himalayan Sheepdog**, **Indian Pariah**, **Jonangi**.

CAT breeds — ① **Long-haired:** **Persian** (brachycephalic; screen for **PKD**), **Himalayan**, **Maine Coon** (screen for **HCM**), **Turkish Angora**, **Ragdoll**. ② **Short-haired:** **Siamese** (points are temperature-sensitive albinism), **Burmese**, **Abyssinian**, **British/American Shorthair**, **Russian Blue**, **Bombay**. ③ **Hairless:** **Sphynx**. **Indian:** **Indian desi/Billi**. **Note:** tortoiseshell/calico = X-linked mosaicism → almost always female.

PEDIGREE SHEET = an official record showing an animal's **ancestry** (usually 3–5 generations) with registration numbers, dates of birth, colour, and the show/working/performance records of the ancestors. Forms: **tabular (bracket)**, **arrow**, and **circular**. **Uses:** **pedigree selection**; **computing F and R** before a mating; **avoiding inbreeding**; breed registration with the **Kennel Club of India (KCI)**; establishing **identity, ownership and sale value**; tracing the origin of a **hereditary defect**. **Selection of a breed** for a client: purpose (guard/companion/working), space & climate, temperament, exercise & grooming need, cost, and the **breed-specific hereditary disorders**.

BREED-LINKED HEREDITARY DEFECTS TO SCREEN

Hip & elbow dysplasia (German Shepherd, Labrador — OFA/PennHIP scoring) · **progressive retinal atrophy** · **von Willebrand's disease** (Doberman) · **congenital deafness** (Dalmatian, white coat) · **brachycephalic airway syndrome** (Pug, Bulldog, Persian) · **PKD** (Persian) · **HCM** (Maine Coon, Ragdoll). Carriers must be identified and excluded from breeding.

23 BREEDING MANAGEMENT — DOGS · CATS · PET BIRDS

Point	DOG (bitch)	CAT (queen)
Puberty	6–12 months	5–9 months
First breeding	2nd–3rd oestrus, ~18–24 months	~10–12 months
Cycle type	MONOESTROUS — ~2 cycles/year, ~6-month interval; spontaneous ovulator	Seasonally polyoestrous , long-day breeder; INDUCED OVULATOR (mating triggers the LH surge)
Oestrus	Proestrus ~9 d (bloody discharge, refuses male) → oestrus ~9 d (accepts male, straw-coloured discharge, flagging)	Oestrus 4–10 d, repeating every 14–21 d in season; calling, lordosis, treading
Best time to mate	Day 10–14 from onset of proestrus ; confirm by vaginal cytology (>80–90% cornified cells) and serum progesterone (~5 ng/mL at the LH surge) ; mate on alternate days	Days 2–4 of oestrus; 2–3 matings to ensure ovulation
Gestation	63 ± 3 days	63–65 days
Litter	1–12 (breed-dependent)	4–6

Management common to both: select on **breed standard conformation, temperament, health screening and pedigree**; **avoid close inbreeding**; deworm and vaccinate before mating; balanced diet with adequate protein, calcium & energy; provide a quiet **whelping/queening box**; monitor parturition (dystocia is common in **brachycephalic** breeds → planned caesarean); record matings, whelping dates and litter details; register the litter.

PET BIRDS COMMON IN INDIA & THEIR BREEDING: **Budgerigar** (clutch 4–6, incubation 18 d), **Cockatiel** (18–21 d), **Lovebirds** (23 d), **Zebra & Bengalese finch**, **Java sparrow**, **Canary**, **fancy pigeons**. Breeding needs: **true pairs** (many species are **monomorphic** → **DNA/endoscopic sexing**), a **nest box** of the right size, 12–14 h light, breeding-season conditioning, **calcium (cuttlefish bone/mineral grit)**, soft/egg food, quiet surroundings; **colony** breeding for finches vs **cage/pair** breeding for budgerigars; avoid inbreeding and colour-mutation matings that carry lethal or semi-lethal genes; foster/hand rearing when needed. **Legal note:** **Indian native birds (rose-ringed parakeet, munias, mynah, peafowl) are protected under the Wildlife (Protection) Act, 1972 and CANNOT be kept, bred or traded** — only exotic species may be kept.

24 WILD ANIMALS — POPULATION DYNAMICS, N E · PLANNED & CONTROLLED BREEDING ★

POPULATION DYNAMICS = study of how a population's size and structure change with time under **natality, mortality, immigration and emigration** ($N_{t+1} = N_t + B - D + I - E$). Growth is **exponential (J-shaped)** when resources are unlimited and **logistic (S-shaped)** when limited by the **carrying capacity K**. Also studied: age structure & life tables, sex ratio, density-dependent (competition, disease) vs density-independent (drought, fire) factors, **Minimum Viable Population** and **Population Viability Analysis (PVA)**.

EFFECTIVE POPULATION SIZE $N_e = 4 N_m N_f / (N_m + N_f)$; $\Delta F = 1 / (2N_e)$

N_e = the size of an **idealised population** that would show the **same rate of inbreeding / loss of genetic variation** as the actual population. N_e is **almost always much SMALLER** than the census number N , and is reduced by: **unequal sex ratio** (a few breeding males), **variance in family size**, **fluctuating population size** (use the **harmonic mean** — a single bottleneck dominates), overlapping generations and non-random mating. **50/500 rule:** $N_e \geq 50$ to **avoid immediate inbreeding depression** ($\Delta F \leq 1\%$ /generation) and $N_e \geq 500$ to **retain long-term evolutionary potential**.

PLANNED BREEDING IN CAPTIVITY: maintained through **STUDBOOKS** (international/national, kept by a studbook keeper) and pedigree analysis. Objectives: **retain ~90% of the founders' gene diversity for 100 years**, **minimise inbreeding**, **equalise founder representation**, **avoid adaptation to captivity**. Pairings are chosen by **MEAN KINSHIP** (mate the lowest-MK animals; **avoid pairs with high kinship**), supported by **genome resource banks** and cooperative programmes (SSP in the USA, EEP in Europe; in India the **Central Zoo Authority conservation-breeding programmes** for the **Red Panda**, **Snow Leopard**, **Sangai (brow-antlered deer)**, **Pygmy Hog**, **Great Indian Bustard**, **Gharial** and **vultures (Jatayu centres)**). Longer generation intervals are **deliberately encouraged** in captive programmes to slow genetic drift — the opposite of livestock practice.

CONTROLLED BREEDING & ASSISTED REPRODUCTION: **oestrus detection** by **non-invasive faecal/urinary steroid monitoring** and **ultrasonography**; **oestrus synchronisation** and **GnRH/hormonal induction**; **semen collection** by **electroejaculation**, evaluation and **cryopreservation in genome resource banks**; **AI** (successful in **elephant, rhino, giant panda, big cats**), **embryo transfer** and **IVF**, **interspecies surrogacy**, and **cloning** for critically endangered species; **contraception (GnRH vaccines, implants)** to prevent surplus animals in over-represented species. **Breeding for conservation** also requires **avoiding hybridisation with related species** and **outbreeding depression**, and planning eventual **reintroduction/rewilding** with disease screening and behavioural preparation.

★ LAST-10-MINUTES: OBJECTIVE AMMUNITION + SUBJECTIVE ATTACK PLANS

Father of Animal Breeding — Jay L. Lush; practice — **Robert Bakewell**

F — Sewall Wright; **Selection index** — Hazel & Lush; **BLUP** — Henderson; **RRS** — Comstock, Hess & Bell; **Contemporary comparison** — Robertson & Rendel; **Heterosis** — Shull

$\Delta G/yr = i \cdot h^2 \cdot \sigma_p / L$; $R = h^2 S$; $i = S/\sigma_p$

L: poultry 1–1.5 · pig 2 · sheep/goat 3–4 · cattle 4.5–6 · **buffalo 6–8 yr**

F full-sib = parent-offspring = 0.25; half-sib 0.125; first cousins 0.0625

R full-sib = parent-offspring = 0.50; half-sib 0.25

Heterozygotes fall by (1 – F); inbreeding changes **genotypic** not gene frequency

Grading up: 50 → 75 → 87.5 → **93.75** → 96.87%; purebred in **4–6 generations**

Heterosis halves in F_2 ; greatest in **low- h^2 fitness traits**

Criss-cross retains ~67%, 3-breed rotation ~86% heterosis

Efficiency: Index > ICL > Tandem

GCA = additive; **SCA = dominance/epistasis**

Exotic inheritance held at 50% (up to 62.5%) in Indian dairy crossbreeding

Buffalo — NO exotic crossbreeding; grade up with **Murrah**

SUBJECTIVE ATTACK PLANS

"Response to selection and factors affecting it" → define selection & S , i → $\Delta G = i \cdot h^2 \cdot \sigma_p / L$ → derive per year → the **8 factors** → species comparison of L →

conclusion on the four levers. **"Inbreeding — coefficient, consequences, applications"** → definition & types → Wright's formula with a worked path → **F** table → genetic consequences (homozygosity, no change in gene frequency) → inbreeding depression → 6 applications → $\Delta F \leq 1\%$ rule. **"Heterosis"** → definition → formula → 3 theories → $H \propto \sum dy^2$ → which traits → types → not fixable → all systems of utilisation → Indian examples. **"Sire evaluation"** → why (sex-limited, AI, 75–90% of gain) → all 7 methods in order → CC in detail → BLUP in detail → accuracy formula → requirements. **"Conservation of AnGR"** → definitions → why → causes of erosion → in-situ vs ex-situ table → NBAGR/DAD-IS/risk status → N_e and the 50/500 rule → Indian programmes.